

Formalization of Post-quantum Reversible Hiding in Encrypted Images from Standard Assumptions

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ABSTRACT

As advancements in computer vision emerge, the necessity for securing image processing techniques also arises, especially in applications such as utilizing federated learning for processing medical images while preserving privacy. Reversible Data Hiding in Encrypted Images (RDHEI) is one of the promising methods that can directly secure visual data while enabling auxiliary information embedding. However, despite extensive research on RDHEI constructions, it has not been formalized as a cryptographic object/primitive with strong security definitions. The absence of such formalization leads to a lack of provable security guarantees and robust privacy comparison between schemes. In this work, we address this gap by presenting the first formal cryptographic treatment of RDHEI. We define the syntax, correctness properties, and security notions, including semantic security, payload privacy, separability, compactness, and robustness. We then construct PQ-RDHEI, an efficient scheme secure against quantum adversaries under the Module Learning With Errors assumption in the quantum random oracle model. The construction employs Module-Lattice Key Encapsulation Mechanism, pseudorandom functions, hash functions, and verifiable secret sharing as building blocks. The experimental evaluation shows PQ-RDHEI achieves higher embedding capacity and stronger encryption entropy in comparison with existing methods while providing formal security guarantees.

1 Introduction

Reversible data hiding (RDH) technique has been the center of attention as it can embed auxiliary information in the cover media and at the same time allow the content's exact recovery¹⁻³. It is this very property that makes RDH ideal for various applications where data and original media preservation are the main concerns, such as medical imaging, military reconnaissance, and legal document archiving⁴⁻⁶. Conventional RDH techniques use the approaches of difference expansion¹, histogram shifting², and prediction error expansion^{3,7} to work on plaintext images. These methods try to find a trade-off between the capacity to embed and the distortion of the cover image.

The rise of cloud computing, along with the growing demand for privacy protection, has led to the need for processing multimedia signals in the encrypted domain^{8,9}; recent developments underscore this evolution, spanning lightweight Feistel-based encryption for IoT security¹⁰; generally, it is common for image owners' systems to encrypt their media before uploading it to cloud storage or processing it in the cloud. Nevertheless, the providers of these services may have to insert metadata into the picture for such purposes as storage management, content annotation, or access control, but they want to carry out this task without decrypting the image. This situation inspired the idea of reversible data hiding in encrypted images (RDHEI) that was first put forward by Puech et al.¹¹ and improved by Zhang¹² later.

RDHEI schemes come under two main categories: proper room imposition after the ciphering (VRAE)¹²⁻¹⁴ and room reservation prior to ciphering (RRBE)¹⁵⁻¹⁷. In the case of VRAE methods, initially, the image is encrypted, and then the space for embedding is created in the encrypted domain. On the other hand, the methods of RRBE provide space for embedding in the plaintext domain prior to encryption and simultaneously exploit spatial correlation in the original image to get a higher capacity¹⁵. Puteaux and Puech¹⁶ introduced a scheme for RRBE that is both fast and is built on the idea of most significant bit (MSB) prediction. This was later enhanced by means of Huffman coding¹⁷ and local difference prediction¹⁸ methods.

To make it more difficult for the attackers to obtain the whole message even if they had various parts of the encrypted image, the secret sharing technique was adopted by various RDHEI schemes in their designs¹⁹⁻²³. These approaches result in the distribution of the encrypted image into many shares so that a threshold number of shares is required for the reconstruction, even if some are defective or not available. Wu et al.¹⁹ were the first to use secret sharing in RDHEI with the difference expansion. Chen et al.²² further allowed multiple data hidings to be supported by this technique. Recently, Saeidi and Mirzakuchaki²⁴ introduced an RDHEI scheme based on Learning With Errors²⁵ (LWE)-based secret sharing utilizing adaptive Huffman coding.

Despite these advances, a fundamental limitation persists: stronger privacy analysis is not possible when the scheme is not formalized, and any security result will remain informal²⁶. The existing RDHEI schemes do not have the necessary rigorous

cryptographic definitions for the primitive itself and its security properties. Claims about confidentiality, integrity, and reversibility cannot be verified using standard cryptographic analysis without formal syntax and security models²⁷. So, despite advances in both steganography and steganalysis, existing RDHEI schemes lack formal cryptographic treatment, which relies instead on heuristic security arguments without rigorous adversarial models or provable guarantees^{28,29}. This situation not only stops meaningful security comparisons but also makes it impossible to use reduction-based proofs that would relate the security of constructions to well-studied computational assumptions.

Moreover, completely encrypted systems and algorithms that are based on classical complexity assumptions might ultimately become obsolete due to the emergence of quantum computers^{30,31}. Recently, the National Institute of Standards and Technology (NIST) has laid down the groundwork for post-quantum cryptographic algorithms³², which in turn makes it imperative to come up with RDHEI schemes that are robust against quantum adversaries.

We present the first formalization of Reversible Data Hiding in Encrypted Images as a cryptographic primitive and introduce security properties for it. To be more precise, our contributions are outlined as follows: First, a formal syntax for RDHEI is provided, which consists of the algorithms (KeyGen, Enc, Embed, Extract, Dec), each with exact input-output specifications. Subsequently, properties of correctness (embedding correctness, decryption correctness) along with security notions (semantic security, payload privacy, robustness, separability, compactness) are formulated using game-based definitions. In the next step, we propose PQ-RDHEI, a post-quantum encrypting scheme, which is not only efficient but also secure against quantum attacks. It is built on ML-KEM³², pseudorandom functions³³, hash functions³⁴, and verifiable secret sharing³⁵. The following step is to show that PQ-RDHEI possesses all the security features specified in the definitions under the Module Learning With Errors (MLWE) assumption in the quantum random oracle model (QROM). Finally, we illustrate via experimental results that PQ-RDHEI provides greater embedding capacity along with encrypted share entropy than the existing best²⁴ techniques.

The remainder of this paper is organized as follows. Section 2 presents the computational assumptions and cryptographic building blocks. In Section 3, we then define the formal syntax and security properties for RDHEI. The subsequent part, Section 4, presents our PQ-RDHEI construction. We provide efficiency analysis and experimental comparison, followed by formal security proofs in Sections 5 and 6. Finally, Section 7 concludes with a discussion of future directions.

2 Preliminaries

We denote by $\lambda \in \mathbb{N}$ the security parameter. For $n \in \mathbb{N}$, we write $[n] := \{1, 2, \dots, n\}$. The notation $\{0, 1\}^\lambda$ denotes the set of λ -bit strings, and $\{0, 1\}^*$ denotes the set of all finite-length binary strings. We write $x \leftarrow \$S$ for uniform random sampling from a set S , and $y \leftarrow \mathcal{A}(x)$ for the output of a deterministic or randomized algorithm $\mathcal{A}$ on input x . The symbol $\|$ denotes string concatenation, and $\oplus$ denotes bitwise XOR. For a string s , we write $|s|$ for its bit-length. A function $f: \mathbb{N} \rightarrow \mathbb{R}$ is *negligible*, written $f(\lambda) = \text{negl}(\lambda)$, if for every polynomial $p(\cdot)$ there exists λ_0 such that $f(\lambda) < 1/p(\lambda)$ for all $\lambda > \lambda_0$. We write PPT for probabilistic polynomial-time and QPT for quantum polynomial-time. The quantum random oracle model (QROM) is the idealized model where all parties have quantum superposition access to a random oracle. IND-CCA denotes indistinguishability under adaptive chosen-ciphertext attack. For images, W and H denote width and height in pixels, respectively.

Computational Assumption

Definition 1 (Module Learning With Errors Problem²⁵). Let λ be a security parameter, n a power of two, q a prime modulus, k the module rank, and $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ the ring of polynomials modulo $X^n + 1$ with coefficients in $\mathbb{Z}_q$. Let χ be an error distribution over R_q (typically a centered binomial distribution β_η parameterized by η). The *Module Learning With Errors* (MLWE _{n,k,q,χ}) problem is defined as follows. Given m samples of the form $(\mathbf{a}_i, b_i) \in R_q^k \times R_q$ where either:

- $\mathbf{a}_i \leftarrow R_q^k$ is uniformly random and $b_i = \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i \in R_q$ for a fixed secret $\mathbf{s} \leftarrow \chi^k$ and error $e_i \leftarrow \chi$ (MLWE distribution), or
- $(\mathbf{a}_i, b_i) \leftarrow R_q^k \times R_q$ are uniformly random (uniform distribution),

distinguish between the two cases with non-negligible advantage. The *decisional MLWE assumption* states that for any PPT adversary $\mathcal{A}$:

$$|\Pr[\mathcal{A}(\{(\mathbf{a}_i, \langle \mathbf{a}_i, \mathbf{s} \rangle + e_i)\}_{i=1}^m) = 1] - \Pr[\mathcal{A}(\{(\mathbf{a}_i, b_i)\}_{i=1}^m) = 1]| = \text{negl}(\lambda),$$

where $\mathbf{s} \leftarrow \chi^k$, $e_i \leftarrow \chi$, $\mathbf{a}_i \leftarrow R_q^k$, and $b_i \leftarrow R_q$ are uniformly random.

Building Blocks

Definition 2 (Pseudorandom Function³³). A pseudorandom function family is a deterministic algorithm $\text{PRF}: \{0, 1\}^\lambda \times \{0, 1\}^* \rightarrow \{0, 1\}^*$ such that for any PPT distinguisher $\mathcal{D}$: $|\Pr_{K \leftarrow \{0, 1\}^\lambda}[\mathcal{D}^{\text{PRF}(K, \cdot)}(1^\lambda) = 1] - \Pr_{f \leftarrow \text{Func}}[\mathcal{D}^{f(\cdot)}(1^\lambda) = 1]| = \text{negl}(\lambda)$, where Func denotes the set of all functions from $\{0, 1\}^*$ to $\{0, 1\}^*$. We instantiate PRF using SHAKE256 as an extendable output function (XOF), which can produce arbitrary-length outputs.

Definition 3 (Hash Function³⁴). A cryptographic hash function $H : \{0,1\}^* \rightarrow \{0,1\}^\lambda$ satisfies collision resistance if for any PPT adversary $\mathcal{A}$: $\Pr[(x,x') \leftarrow \mathcal{A}(1^\lambda) : x \neq x' \wedge H(x) = H(x')] = \text{negl}(\lambda)$. We instantiate H using SHA3-256, which provides post-quantum collision resistance.

Definition 4 (Module Lattice Key Encapsulation Mechanism³²). The key encapsulation mechanism ML-KEM, based on MLWE problem, with parameters $(n,k,q,\eta_1,\eta_2,d_u,d_v)$ consists of ML-KEM = (KeyGen, Encaps, Decaps), where $\text{Compress}_q(x,d) = \lfloor (2^d/q) \cdot x \rfloor \bmod 2^d$ and $\text{Decompress}_q(x,d) = \lfloor (q/2^d) \cdot x \rfloor$ perform lossy compression and decompression to d bits.

- $(pk,sk) \leftarrow \text{KeyGen}(1^\lambda)$: On input the security parameter 1^λ , the key generation algorithm samples a seed $\rho \leftarrow \{0,1\}^{256}$, generates matrix $\hat{A} \in R_q^{k \times k}$ from ρ using XOF, samples $\mathbf{s}, \mathbf{e} \leftarrow \beta_\eta^k$ from centered binomial distribution β_η , computes $\mathbf{t} = \hat{A}\mathbf{s} + \mathbf{e} \in R_q^k$, and outputs public key $pk = (\mathbf{t}, \rho)$ and secret key $sk = \mathbf{s}$.
- $(c,K) \leftarrow \text{Encaps}(pk)$: On input a public key pk , the encapsulation algorithm samples $m \leftarrow \{0,1\}^{256}$ uniformly, computes $(K',r) = G(H(pk) \| m)$ where G is a key derivation function, samples $\mathbf{r}', \mathbf{e}_1, \mathbf{e}_2$ from β_η using r as seed, computes $\mathbf{u} = \hat{A}^T \mathbf{r}' + \mathbf{e}_1 \in R_q^k$ and $v = \mathbf{t}^T \mathbf{r}' + \mathbf{e}_2 + \lfloor q/2 \rfloor \cdot \text{Decode}(m) \in R_q$, compresses to $c = (\text{Compress}_q(\mathbf{u}, d_u), \text{Compress}_q(v, d_v))$, and outputs shared key $K = \text{KDF}(K' \| H(c))$ and ciphertext c .
- $K \leftarrow \text{Decaps}(sk, c)$: On input a secret key sk and ciphertext c , the decapsulation algorithm parses $c = (\mathbf{u}', v')$, decompresses to $\mathbf{u} = \text{Decompress}_q(\mathbf{u}', d_u)$ and $v = \text{Decompress}_q(v', d_v)$, computes $m' = \text{Encode}(\lfloor (v - \mathbf{s}^T \mathbf{u}) \cdot 2/q \rfloor \bmod 2)$, recomputes $(K', r) = G(H(pk) \| m')$, verifies encapsulation by recomputing c' with (pk, m', r) , and outputs $K = \text{KDF}(K' \| H(c))$ if $c = c'$, otherwise outputs a pseudorandom key derived from a secret rejection value.

Definition 5 (Verifiable Secret Sharing³⁵). The hash-based verifiable secret sharing scheme Π_{LA} , operating in the quantum random oracle model with threshold (t,n) where $n \geq 2t+1$, consists of $\Pi_{\text{LA}} = (\text{Share}, \text{VerifyShare}, \text{Reconstruct})$.

- $(\{s_i, \gamma_i\}_{i=1}^n, \pi_{\text{share}}) \leftarrow \text{Share}(s, t, n)$: On input secret $s \in \mathbb{Z}_N$, threshold t , and number of parties n , the sharing algorithm samples random degree- t polynomials $f(X), r(X) \in \mathbb{Z}_N[X]_t$ with $f(0) = s$, computes shares $s_i = f(i)$ and randomizers $\gamma_i = r(i)$ for $i \in [1, n]$, generates commitments $c_i = H(s_i \| \gamma_i)$, computes challenge $d = H(c_1 \| \dots \| c_n)$, sets $z(X) = r(X) + d \cdot f(X)$, and outputs shares $\{s_i, \gamma_i\}_{i=1}^n$ and proof $\pi_{\text{share}} = (z(X), c_1, \dots, c_n)$.
- $\{0,1\} \leftarrow \text{VerifyShare}(i, s_i, \gamma_i, \pi_{\text{share}})$: On input index i , share s_i , randomizer γ_i , and $\pi_{\text{share}} = (z(X), c_1, \dots, c_n)$, the verification algorithm checks if $\deg(z(X)) \leq t$, computes $d = H(c_1 \| \dots \| c_n)$, verifies $c_i = H(s_i \| z(i) - d \cdot s_i)$, and outputs 1 if all checks pass, else 0.
- $s \leftarrow \text{Reconstruct}(\{(i, s_i, \gamma_i)\}_{i \in S}, \pi_{\text{share}})$: On input a set of verified share tuples $\{(i, s_i, \gamma_i)\}_{i \in S}$ where $|S| \geq t+1$ and π_{share} , the reconstruction algorithm verifies each share using VerifyShare, computes Lagrange coefficients $\lambda_i^S = \prod_{j \in S, j \neq i} \frac{j}{j-i} \pmod{N}$, reconstructs $s = \sum_{i \in S} \lambda_i^S \cdot s_i \pmod{N}$, and outputs s .

In our PQ-RDHEI in the next sections, ML-KEM is for key encapsulation to establish shared encryption keys, relying on MLWE hardness with the content owner holding the public key and secret key for decryption; SHAKE256 as a pseudorandom function for block mask and payload masking generation, with keys derived from the KEM shared secret; SHA3 for key derivation, commitment, and tag generation, assuming collision resistance in QROM without trusted setup; secret sharing Π_{LA} for payload integrity verification, relying on binding properties in QROM where dealers generate shares and any $t+1$ parties reconstruct; Fisher-Yates permutation (Definition 14) with PRF-seeded deterministic shuffle for block position randomization, recoverable with the encryption key; and Huffman coding (Remark 2) with adaptive codebook for MSB compression to increase embedding capacity, which provides information-theoretic guarantees with deterministic codebooks from block statistics.

As for trust assumptions, the content owner holds (ek, dk) . The data hider receives (hk, xk) and can embed/extract without accessing plaintext. Under the standard key model $(hk = xk)$, both operations are available to the same party. The QROM assumption models hash functions as quantum-accessible random oracles, which reflects the post-quantum security goal.

3 Syntax for Reversible Hiding in Encrypted Images

We treat RDHEI as a cryptographic construction by specifying its syntax through a tuple of polynomial-time algorithms, defining correctness via probabilistic guarantees, and formulating security properties through adversarial games in the following.

Definition 6 (RDHEI). Let $\mathcal{M}$ be a message space, $\mathcal{D}$ a payload space, and $\kappa : \mathbb{N} \rightarrow \mathbb{N}$ a capacity function. A *reversible data hiding in encrypted domain* scheme is a tuple of PPT algorithms $\text{RDHED} = (\text{KeyGen}, \text{Enc}, \text{Embed}, \text{Extract}, \text{Dec})$ defined as follows:

- $\text{KeyGen}(1^\lambda) \rightarrow (ek, dk, hk, xk)$: On input security parameter λ , outputs encryption key ek , decryption key dk , hiding key hk , and extraction key xk .
- $\text{Enc}(ek, m) \rightarrow c$: On input ek and $m \in \mathcal{M}$, outputs ciphertext c .

- $\text{Embed}(hk, c, d) \rightarrow \hat{c}$: On input hk , ciphertext c , and payload $d \in \mathcal{D}$ with $|d| \leq \kappa(|m|)$, outputs marked ciphertext $\hat{c}$.
- $\text{Extract}(xk, \hat{c}) \rightarrow (c, d)$: On input xk and marked ciphertext $\hat{c}$, outputs ciphertext c and payload d , or $\perp$.
- $\text{Dec}(dk, c) \rightarrow m$: On input dk and ciphertext c , outputs $m \in \mathcal{M}$ or $\perp$.

Remark 1 (Key Distribution Model). The keys (ek, dk) are held by the content owner; (hk, xk) are held by the data hider. A receiver may possess any subset. If ek is public, the scheme supports public-key encryption.

Definition 7 (Correctness). An RDHEI scheme satisfies *embedding correctness* if for all $\lambda, m \in \mathcal{M}, d \in \mathcal{D}$ with $|d| \leq \kappa(|m|)$:

$$\Pr[(ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda); c \leftarrow \text{Enc}(ek, m); \hat{c} \leftarrow \text{Embed}(hk, c, d); (c', d') \leftarrow \text{Extract}(xk, \hat{c}); c' = c \wedge d' = d] = 1 - \text{negl}(\lambda).$$

Definition 8 (Decryption Correctness). An RDHEI scheme satisfies *decryption correctness* if for all $\lambda, m \in \mathcal{M}$:

$$\Pr[(ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda); c \leftarrow \text{Enc}(ek, m); \text{Dec}(dk, c) = m] = 1 - \text{negl}(\lambda).$$

Definition 9 (Semantic Security). An RDHEI scheme satisfies *semantic security* if for any PPT adversary $\mathcal{A}$, the advantage $\text{Adv}_{\mathcal{A}}^{\text{sem}}(\lambda) = |\Pr[\text{Expt}_{\mathcal{A},0}^{\text{sem}}(\lambda) = 1] - \Pr[\text{Expt}_{\mathcal{A},1}^{\text{sem}}(\lambda) = 1]|$ is negligible, where $\text{Expt}_{\mathcal{A},b}^{\text{sem}}(\lambda)$ is defined as: $(ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda)$. Give (ek, hk, xk) to $\mathcal{A}$. $\mathcal{A}$ outputs (m_0, m_1) with $|m_0| = |m_1|$. Challenger computes $c^* \leftarrow \text{Enc}(ek, m_b)$ and gives c^* to $\mathcal{A}$. $\mathcal{A}$ outputs bit b' . Output b' .

Definition 10 (Payload Privacy). An RDHEI scheme satisfies *payload privacy* if for any PPT adversary $\mathcal{A}$ we have $\text{Adv}_{\mathcal{A}}^{\text{pay}}(\lambda) = |\Pr[\text{Expt}_{\mathcal{A},0}^{\text{pay}}(\lambda) = 1] - \Pr[\text{Expt}_{\mathcal{A},1}^{\text{pay}}(\lambda) = 1]| = \text{negl}(\lambda)$, where $\text{Expt}_{\mathcal{A},b}^{\text{pay}}(\lambda)$ is defined by the following procedure: $(ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda)$. Give (ek, hk) to $\mathcal{A}$. $\mathcal{A}$ outputs (m, d_0, d_1) with $|d_0| = |d_1| \leq \kappa(|m|)$. $c \leftarrow \text{Enc}(ek, m)$; $\hat{c}^* \leftarrow \text{Embed}(hk, c, d_b)$. Give $\hat{c}^*$ to $\mathcal{A}$. $\mathcal{A}$ outputs b' . Output b' .

Definition 11 (Separability). An RDHEI scheme is *separable* if: (I) $\text{Extract}(xk, \cdot)$ can be computed without knowledge of dk (extraction-first), and (II) there exists $\text{Dec}'(dk, \hat{c}) \rightarrow \hat{m}$ computable without xk , where $\hat{m}$ encodes both m and d in a recoverable manner (decryption-first).

Definition 12 (Compactness). An RDHEI scheme is *compact* if there exists a polynomial $p(\cdot)$ such that for all valid (c, d) we have $|\text{Embed}(hk, c, d)| \leq |c| + p(\lambda, |d|)$. The scheme is *strongly compact* if $|\text{Embed}(hk, c, d)| = |c|$.

Definition 13 (Robustness). An RDHEI scheme satisfies *embedding robustness* if for any PPT adversary $\mathcal{A}$:

$$\Pr[(ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda); (c, d, \hat{c}^*) \leftarrow \mathcal{A}(ek, dk, hk, xk); (c', d') \leftarrow \text{Extract}(xk, \hat{c}^*); \hat{c}^* \neq \text{Embed}(hk, c, d); \wedge c' = c \wedge d' = d] = \text{negl}(\lambda).$$

Prior RDHEI schemes operated under informal security assumptions. We contextualize our formalized definitions:

- Existing works^{12,16,24} use XOR-based ciphers or chaotic maps, assuming computational indistinguishability without formal reduction. Our semantic security (Definition 9) provides a game-based formulation reducible to IND-CCA security of the underlying KEM.
- Prior schemes^{19,22} protect payloads via embedding location obscurity rather than cryptographic guarantees. Our payload privacy (Definition 10) ensures that adversaries knowing the embedding algorithm cannot distinguish between payloads.
- Most RDHEI constructions lack integrity verification. Our robustness definition (Definition 13) ensures adversarially-crafted ciphertexts cannot be misattributed to honest embeddings, protecting against tampering attacks.
- The separability notion (Definition 11) formalizes extraction-first and decryption-first capabilities that prior works^{14,21} describe informally, enabling detailed comparison of operational modes.

4 Proposed Formalized RDHEI Scheme

We now present our post-quantum RDHEI construction, beginning with auxiliary definitions, followed by the main design and its instantiation from the building blocks introduced in Section 2.

Definition 14 (Fisher-Yates Permutation³⁶). The deterministic Fisher-Yates algorithm $\text{FisherYates}(\ell, K)$ generates a pseudorandom permutation $\sigma: [\ell] \rightarrow [\ell]$ as follows: initialize array $A = [1, 2, \dots, \ell]$; for $i = \ell$ down to 2: compute $j = (\text{PRF}(K, i) \bmod i) + 1$, swap $A[i] \leftrightarrow A[j]$; output permutation σ where $\sigma(k) = A[k]$.

Definition 15 (Embedding Capacity). The capacity function $\kappa: \mathbb{N} \rightarrow \mathbb{N}$ for PQ-RDHED with parameters (n_b, β, T) on image of size $W \times H$ is defined as: $\kappa(W \cdot H) = \sum_{k \in \mathcal{E}} \left(\beta \cdot n_b^2 - \sum_{u=1}^{n_b^2} |\text{cw}_{k,u}| \right) - |\text{overhead}|$, where $\mathcal{E}$ is the set of embeddable blocks determined by threshold T , and $|\text{overhead}|$ accounts for the length encoding, auxiliary data, and verification proof. The expected capacity satisfies $\kappa(W \cdot H) = \Theta(W \cdot H \cdot \beta / n_b^2)$ for typical natural images.

Remark 2 (Adaptive Huffman Codebook). For embeddable blocks with $\alpha \in \{1, 2, 3, 4\}$ distinct MSB types, the Huffman codebook is predefined as follows:

α	1	2	3	4
MSB Rank	1	1, 2	1, 2, 3	1, 2, 3, 4
Codeword	1	1, 0	1, 01, 00	1, 01, 001, 000

The encoded result for pixel u in block k is $(0)^{p_{k,u}} \| 1 \| \overline{\text{code}}$, where $p_{k,u}$ counts prefix zeros allocated for embedding and $\overline{\text{code}}$ is the reversed Huffman codeword.

We now present our main construction. Let $\mathcal{M} = \{0, 1, \dots, 255\}^{W \times H}$ denote the message space of grayscale images with dimensions $W \times H$, and let $\mathcal{D} = \{0, 1\}^*$ denote the payload space. We define the block size parameter $n_b \in \mathbb{N}$ (typically $n_b = 4$ or $n_b = 8$), the MSB parameter $\beta \in \{3, 4, 5, 6\}$ denoting the number of most significant bits analyzed, and the embeddability threshold $T \in \{2, 3, 4\}$.

Remark 3. Our construction employs a hybrid KEM-DEM (Key Encapsulation Mechanism - Data Encapsulation Mechanism) paradigm for image encryption. The ML-KEM component provides post-quantum secure key establishment, outputting a shared secret K . The DEM component is instantiated as a PRF-based stream cipher operating in counter mode: for each block index k , the encryption key K_e (derived from K) seeds the PRF to generate an 8-bit block mask $\mu_k = \text{PRF}(K_e, k)[0:8]$, which is XORed with each pixel in block k . This design ensures that pixels within the same block share a common mask, thereby preserving intra-block MSB correlation structure essential for identifying embeddable regions during the data hiding phase.

To specify concrete parameters for PQ-RDHED instantiation and analyze security levels, we instantiate ML-KEM at security level 3, providing approximately 192-bit classical and 128-bit quantum security³² with parameters $n = 256$, $k = 3$, $q = 3329$, $\eta_1 = \eta_2 = 2$, $d_u = 10$, $d_v = 4$, yielding key and ciphertext sizes $|pk| = 1184$ bytes, $|sk| = 2400$ bytes, $|c_{\text{kem}}| = 1088$ bytes, and $|K| = 32$ bytes. We set the parameter $\lambda = 256$ bits, and SHA3 provides 256-bit classical and 128-bit quantum collision resistance under Grover's algorithm³⁷. SHAKE256 as an XOF provides PRF security with adaptable output length. Image processing parameters are: block size $n_b \in \{2, 4, 8\}$, MSB bits per pixel $\beta \in \{3, 4, 5, 6\}$, and embeddability threshold $T \in \{2, 3, 4\}$. For high-capacity embedding, we recommend $(n_b, \beta, T) = (4, 5, 4)$; for higher fidelity, $(n_b, \beta, T) = (2, 4, 2)$ provides more granular control. Under these parameters, the security advantages satisfy $\text{Adv}_{\mathcal{A}}^{\text{sem}}(\lambda) \leq 2^{-128} + \ell \cdot 2^{-256} \approx 2^{-128}$ and $\text{Adv}_{\mathcal{A}}^{\text{robust}}(\lambda) \leq 2^{-128} + 2^{-128} + q_H^2 \cdot 2^{-256} \approx 2^{-127}$ where $\ell \leq 2^{20}$ is the maximum number of blocks (supporting images up to 4096×4096 with $n_b = 4$) and $q_H \leq 2^{64}$ is the maximum number of hash queries, which ensures security margins well above the 100-bit threshold for long-term security.

Definition 16 (Post-Quantum RDHEI). The post-quantum reversible data hiding in encrypted domain scheme PQ-RDHED, instantiated from ML-KEM, PRF, H, and Π_{LA} with parameters (λ, n_b, β, T) , consists of the tuple PQ-RDHED = (KeyGen, Enc, Embed, Extract, Dec) as formally specified in Figure 1. The message space is $\mathcal{M} = \{0, 1, \dots, 255\}^{W \times H}$ for grayscale images of dimensions $W \times H$, the payload space is $\mathcal{D} = \{0, 1\}^*$, and the capacity function κ is defined per Definition 15.

The scheme operates through a key generation, where ML-KEM.KeyGen establishes the asymmetric encryption infrastructure (pk, sk) while independent λ -bit seeds K_h and K_x are sampled for the hiding and extraction keys respectively. During encryption, the input image m undergoes spatial partitioning via $\text{Partition}(\cdot, n_b)$, which segments the pixel array into $\ell = \lceil W/n_b \rceil \cdot \lceil H/n_b \rceil$ non-overlapping blocks of size $n_b \times n_b$. The KEM-DEM paradigm then proceeds: ML-KEM.Encaps produces encapsulation c_{kem} and shared secret K , from which domain-separated keys K_e and K_π are derived for block-wise stream cipher encryption and Fisher-Yates permutation, respectively. Each block receives a common 8-bit mask $\mu_k = \text{PRF}(K_e, k)[0:8]$, preserving intra-block MSB correlation essential for subsequent embedding.

The embedding phase initiates with MSB analysis: for each encrypted block, the procedure extracts the β most significant bits of each pixel and counts distinct values via the set cardinality α_k . Blocks satisfying $\alpha_k \leq T$ are deemed embeddable and recorded in the location map L . The function SortByFreq orders the α_k distinct MSB values by occurrence frequency, enabling optimal Huffman code assignment per Remark 2. The auxiliary function $\text{WriteEncoded}(E'_k, \{\text{cw}_{k,u}\}_u, \text{emb}_k, \beta)$ then performs an interleaving operation: it replaces each pixel's β -MSB with the concatenation of embedding bits (placed in prefix zero positions), a delimiter bit, and the reversed Huffman codeword. The complete embedding stream—comprising length-prefixed masked payload $\tilde{d}$, auxiliary metadata aux , and VSS share π_{share} —is distributed across embeddable blocks in lexicographic order, culminating in a commitment tag tag binding the hiding key to the embedded content.

PQ-RDHEI.KeyGen(1^λ)	PQ-RDHEI.Embed(hk, c, d)	PQ-RDHEI.Extract($\hat{x}, \hat{c}$)	PQ-RDHEI.Dec(dk, c)
<pre>1: $(pk, sk) \leftarrow \text{ML-KEM.KeyGen}(1^\lambda)$ 2: $K_h \leftarrow \mathcal{H}(0, 1)^\lambda$ 3: $K_s \leftarrow K_h$ 4: $ek \leftarrow pk; \quad dk \leftarrow sk$ 5: $hk \leftarrow K_h; \quad xk \leftarrow K_s$ 6: return (ek, dk, hk, xk)</pre>	<pre>1: $(c_{\text{kem}}, \{E'_k\}_{k=1}^\ell) \leftarrow c$ 2: $K_{\text{pos}} \leftarrow \text{H}(hk \ 0x02); \quad K_{\text{mask}} \leftarrow \text{H}(hk \ 0x03)$ 3: $L \leftarrow \emptyset; \quad \mathcal{E} \leftarrow \emptyset$ 4: for $k \in [\ell]$ do 5: for $u \in [n_b^2]$ do $\text{msb}_{k,u} \leftarrow \lfloor e_{k,u} / 2^{8-\beta} \rfloor$ 6: $\alpha_k \leftarrow \lfloor \lfloor \text{msb}_{k,u}; u \in [n_b^2] \rfloor \rfloor$ 7: if $\alpha_k \leq T$ then $\mathcal{E} \leftarrow \mathcal{E} \cup \{k\}; \quad L[k] \leftarrow 1$ 8: for $k \in \mathcal{E}$ do 9: $(\text{msb}_k^{(1)}, \dots, \text{msb}_k^{(\alpha_k)}) \leftarrow \text{SortByFreq}(\{\text{msb}_{k,u}\}_u)$ 10: $\text{code}_k \leftarrow \text{Huffman}(\alpha_k)$ 11: for $u \in [n_b^2]$ do 12: $\text{cw}_{k,u} \leftarrow (0)^{p_{k,u}} \ 1 \ \text{code}_k(\text{msb}_{k,u})$ 13: $\text{cap}_k \leftarrow \beta \cdot n_b^2 - \sum_{u=1}^{n_b^2} \text{cw}_{k,u}$ 14: $\text{aux} \leftarrow L \ \mathcal{E} \ \{(\alpha_k, \{\text{msb}_k^{(j)}\}_{j=1}^{\alpha_k})\}_{k \in \mathcal{E}}$ 15: $(\{s_i, \gamma_i\}_{i=1}^2, \pi_{\text{share}}) \leftarrow \Pi_{\text{LA}}.\text{Share}(\text{H}(\text{aux} \ d), 1, 2)$ 16: $\hat{d} \leftarrow d \oplus \text{PRF}(K_{\text{mask}}, 0x00) \ 0 : d$ 17: $\text{stream} \leftarrow [d]_{32} \ \hat{d} \ \text{aux} \ \pi_{\text{share}}$ 18: $\text{ptr} \leftarrow 0$ 19: for $k \in \mathcal{E}$ do 20: $\text{emb}_k \leftarrow \text{stream}[\text{ptr} : \text{ptr} + \text{cap}_k]$ 21: $\text{ptr} \leftarrow \text{ptr} + \text{cap}_k$ 22: $\hat{E}_k \leftarrow \text{WriteEncoded}(E'_k, \{\text{cw}_{k,u}\}_u, \text{emb}_k, \beta)$ 23: for $k \in [\ell] \setminus \mathcal{E}$ do $\hat{E}_k \leftarrow E'_k$ 24: $\text{tag} \leftarrow \text{H}(hk \ c_{\text{kem}} \ \text{H}(\text{stream}))$ 25: $\hat{c} \leftarrow (c_{\text{kem}}, \{\hat{E}_k\}_{k=1}^\ell, \text{tag})$ 26: return $\hat{c}$</pre>	<pre>1: $(c_{\text{kem}}, \{E'_k\}_{k=1}^\ell) \leftarrow \hat{c}$ 2: $K \leftarrow \text{ML-KEM.Decaps}(dk, c_{\text{kem}})$ 3: $K_e \leftarrow \text{H}(K \ 0x00); \quad K_s \leftarrow \text{H}(K \ 0x01)$ 4: $\sigma \leftarrow \text{FisherYates}(\ell, K_s); \quad \sigma^{-1} \leftarrow \text{Invert}(\sigma)$ 5: for $k \in [\ell]$ do $E_k \leftarrow E'_{\sigma^{-1}(k)}$ 6: for $k \in [\ell]$ do 7: $\mu_k \leftarrow \text{PRF}(K_e, k) \ 0 : 8$ 8: for $u \in [n_b^2]$ do 9: $p_{k,u} \leftarrow e_{k,u} \oplus \mu_k$ 10: $B_k \leftarrow \{p_{k,u}\}_{u \in [n_b^2]}$ 11: $m \leftarrow \text{Reconstruct}(\{B_k\}_{k=1}^\ell, W, H, n_b)$ 12: return m</pre>	
PQ-RDHEI.Enc(ek, m)			
<pre>1: $\{p_{i,j}\}_{i \in [W], j \in [H]} \leftarrow m$ 2: $\ell \leftarrow \lceil W/n_b \rceil \cdot \lceil H/n_b \rceil$ 3: $\{B_k\}_{k=1}^\ell \leftarrow \text{Partition}(\{p_{i,j}\}, n_b)$ 4: $(c_{\text{kem}}, K) \leftarrow \text{ML-KEM.Encaps}(ek)$ 5: $K_e \leftarrow \text{H}(K \ 0x00); \quad K_s \leftarrow \text{H}(K \ 0x01)$ 6: for $k \in [\ell]$ do 7: $\mu_k \leftarrow \text{PRF}(K_e, k) \ 0 : 8$ 8: for $u \in [n_b^2]$ do 9: $e_{k,u} \leftarrow p_{k,u} \oplus \mu_k$ 10: $\hat{E}_k \leftarrow \{e_{k,u}\}_{u \in [n_b^2]}$ 11: $\sigma \leftarrow \text{FisherYates}(\ell, K_s)$ 12: for $k \in [\ell]$ do $E_k \leftarrow E_{\sigma(k)} \leftarrow \hat{E}_k$ 13: $c \leftarrow (c_{\text{kem}}, \{E'_k\}_{k=1}^\ell)$ 14: return c</pre>			

Figure 1. Our Post-Quantum Reversible Data Hiding in Encrypted Image PQ-RDHEI = (KeyGen, Enc, Embed, Extract, Dec) based upon the standard building blocks Key Encapsulation ML-KEM = (KeyGen, Encaps, Decaps), Pseudorandom Function PRF, Hash Function $\mathcal{H}$, and Verifiable Secret Sharing Π_{LA} = (Share, VerifyShare, Reconstruct) with parameters (n_b, β, T) .

Extraction reverses this process through pattern detection: $\text{DetectPattern}(\hat{E}_k, \beta)$ identifies embeddable blocks by scanning for the characteristic prefix-zero-delimiter structure in MSB positions. The function $\text{ParseMSB}(\hat{E}_k[u], \beta)$ decomposes each pixel's upper bits into embedding prefix length $p_{k,u}$ and encoded codeword, while Huffman^{-1} recovers original MSB values via reverse Huffman decoding. After stream reassembly and VSS verification, $\text{RestoreMSB}(\hat{E}_k, \{\text{msb}_{k,u}\}_u, \beta)$ reconstructs the original encrypted blocks by reinstating decoded MSB values into their canonical positions. Decryption then proceeds conventionally: KEM decapsulation recovers K , inverse permutation σ^{-1} (computed via $\text{Invert}(\sigma)$) restores block ordering, XOR with regenerated masks yields plaintext blocks, and finally $\text{Reconstruct}(\{B_k\}_{k=1}^\ell, W, H, n_b)$ reassembles the original image from its constituent blocks.

For images where dimensions are not divisible by n_b , boundary blocks are zero-padded with original dimensions stored in aux for exact recovery. The Extract algorithm returns $\perp$ upon tag verification failure, VSS verification failure, or malformed stream parsing, without leaking failure-mode information.

5 PQ-RDHED Comparison and Efficiency Analysis

It has been shown that Saeidi & Mirzakuchaki's technique is the most efficient RDHEI²⁴, so if our method is more efficient than Saeidi & Mirzakuchaki's technique, then our work is the most efficient RDHEI; thus, in the following, we focus on comparing our technique with Saeidi & Mirzakuchaki's method. We set up our experiments with parameters and specifications of Saeidi & Mirzakuchaki to facilitate an accurate comparison.

From an embedding capacity perspective, PQ-RDHEI has better performance across all test images. At the same time, both methods achieve equivalent perfect reversibility. However, PQ-RDHEI additionally exhibits higher encrypted share entropy, indicating stronger encryption properties in comparison with classical RDHEI.

The comparative analysis in Table 1(a) reveals distinct embedding capacity characteristics between the two methods. For the parameter configuration $\beta = 5, n_b = 4, T = 4$ in PQ-RDHEI versus $\beta = 5, n = 8, T = 4$ in Saeidi & Mirzakuchaki, the pure capacity ratios range from a factor of 17.6 for Airplane (595,728 bits versus 33,786 bits) to 4.8 for Baboon (160,190 bits versus 33,896 bits). This variation correlates with image texture complexity, as images with smoother regions (Airplane, Peppers, Jet) admit substantially higher capacities under the $n_b = 4$ block structure, whereas highly textured images (Baboon) exhibit reduced embeddable block counts. The embedding rate metric quantifies this precisely: PQ-RDHEI achieves 2.2725 bits per pixel for Airplane compared to 0.1289 bpp in the reference method, while Baboon yields 0.6111 bpp versus 0.1293 bpp. The entropy analysis in Table 2(b) demonstrates that PQ-RDHEI encrypted shares attain entropy values approximating the theoretical

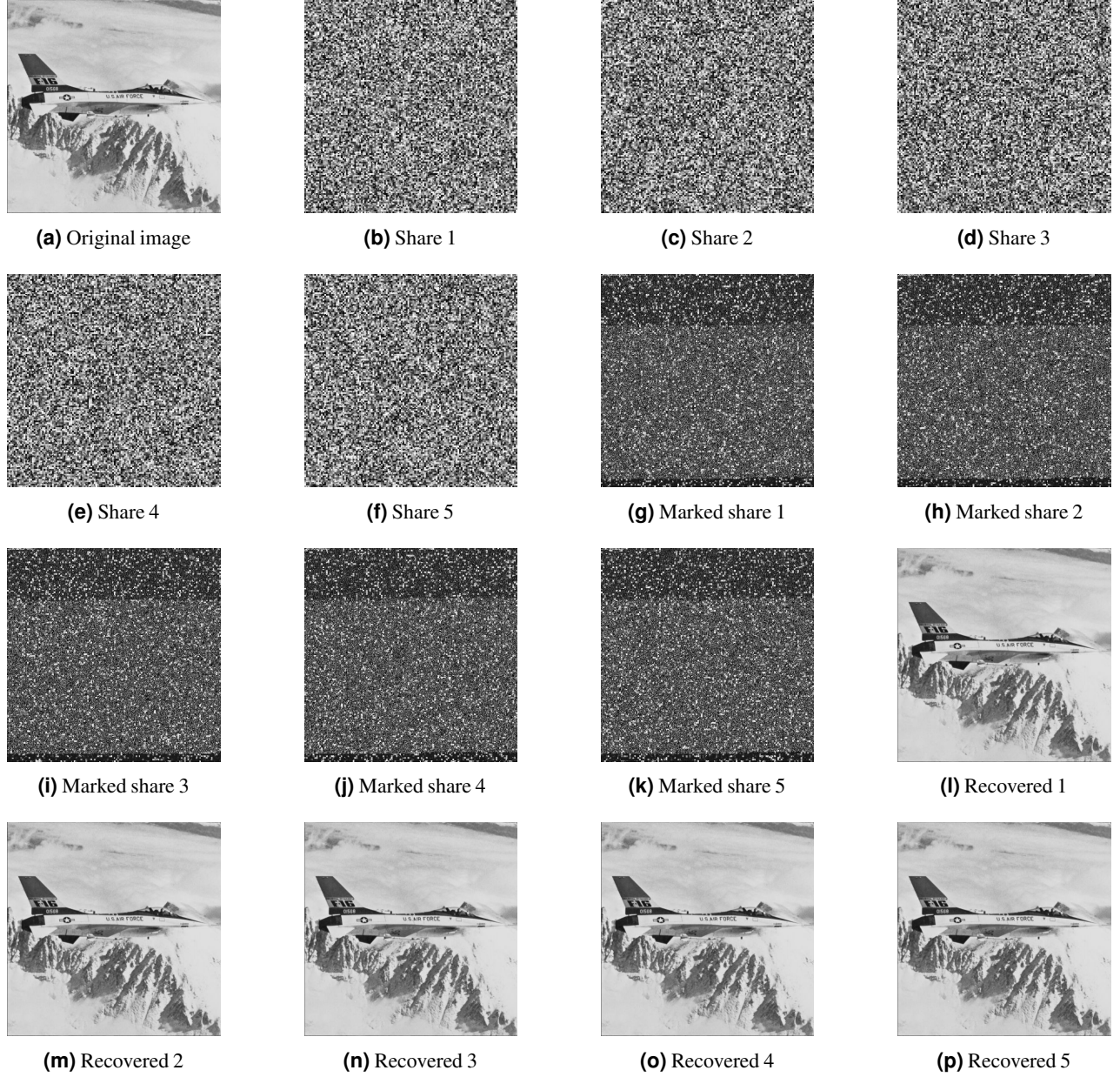


Figure 2. Demonstration of our proposed Post-quantum RDHEI technique using the Jet sample image. (a) The original cover image. (b–f) Five encrypted shares generated via ML-KEM-based encryption, each exhibiting noise-like appearance with no discernible visual information. (g–k) Marked encrypted shares after data embedding, maintaining indistinguishability from unmodified encrypted shares. (l–p) Recovered images obtained by decrypting any four of the five marked shares, demonstrating perfect reconstruction with $\text{PSNR} = +\infty$ and $\text{SSIM} = 1.0$, confirming lossless reversibility of the scheme.

maximum of 8 bits (ranging from 7.9902 to 7.9992), whereas the reference method produces shares with entropy near 7.48 bits. This 0.5-bit entropy increase per pixel indicates enhanced statistical randomness in the encrypted domain, directly attributable to the ML-KEM-based KEM-DEM encryption structure versus the stream cipher-only approach.

Table 2a and Table 3 quantify the embedding rate dependence on block geometry and MSB parameter selection. For 2×2 blocks with $\beta = 5$, $T = 2$, PQ-RDHEI embedding rates span 0.9295 bpp (Baboon) to 2.7906 bpp (Airplane), contrasting with the reference range of 0.9590 bpp to 1.7682 bpp under comparable conditions. The 4×4 block configuration reduces maximum rates to 0.1298 bpp (Baboon, $\beta = 5$, $T = 2$) through 1.9287 bpp (Peppers, $\beta = 5$, $T = 3$) for PQ-RDHEI, compared to 0.3589 bpp through 0.4160 bpp in the reference method. This block size sensitivity reflects the fundamental trade-off between spatial correlation preservation and embeddable region identification. The histogram analysis in Figure 3 provides distributional

validation: pixel value distributions in recovered images exhibit exact correspondence with original distributions, confirming bit-exact reconstruction. The concentration of histogram mass at intensity values 123, 127, 120, and 125 in the Airplane example persists unchanged through the encryption-embedding-extraction-decryption pipeline, verifying that the reversibility property holds at the pixel-level distribution granularity rather than merely at aggregate statistical measures.

Table 1. Performance comparison with Saeidi & Mirzakuchaki²⁴

(a) Embedding capacity comparison ($\beta=5$, $n=8$, $T=4$ vs. $\beta=5$, $n_b=4$, $T=4$)

Sample Image	Saeidi & Mirzakuchaki ²⁴			Our PQ-RDHEI		
	Entire Capacity (bits)	Pure Capacity (bits)	Embedding Rate (bpp)	Entire Capacity (bits)	Pure Capacity (bits)	Embedding Rate (bpp)
Airplane	489962	33786	0.1289	713955	595728	2.2725
Baboon	490680	33896	0.1293	160190	160190	0.6111
Barbara	517601	35899	0.1369	588832	588832	1.4198
Boat	262144	11263	0.0430	424278	424278	1.6185
Cameraman	726708	46269	0.1765	121155	121155	1.8487
Jet	664582	45652	0.1741	547026	531712	2.0283
Lake	489669	33791	0.1289	407241	407241	1.5535
Peppers	524288	23551	0.0898	573155	573155	2.1864

(b) PSNR and SSIM of recovered images

Sample Image	Saeidi & Mirzakuchaki ²⁴		Our PQ-RDHEI	
	PSNR (dB)	SSIM	PSNR (dB)	SSIM
Airplane	$+\infty$	1.000000	$+\infty$	1.000000
Baboon	$+\infty$	1.000000	$+\infty$	1.000000
Barbara	$+\infty$	1.000000	$+\infty$	1.000000
Boat	$+\infty$	1.000000	$+\infty$	1.000000
Cameraman	$+\infty$	1.000000	$+\infty$	1.000000
Jet	$+\infty$	1.000000	$+\infty$	1.000000
Lake	$+\infty$	1.000000	$+\infty$	1.000000
Peppers	$+\infty$	1.000000	$+\infty$	1.000000

Table 2. Comparison with Saeidi & Mirzakuchaki²⁴ for block size 2×2 and entropy analysis ($\beta=5$, $T=4$)

(a) Embedding rates (bpp) for block size 2×2

Image	Saeidi & Mirzakuchaki ²⁴				Our PQ-RDHEI			
	$\beta=3$, $T=2$	$\beta=4$, $T=2$	$\beta=5$, $T=2$	$\beta=6$, $T=2$	$\beta=3$, $T=2$	$\beta=4$, $T=2$	$\beta=5$, $T=2$	$\beta=6$, $T=2$
Airplane	1.6845	1.9937	0.9590	1.0300	0.9888	1.9359	2.7906	3.4341
Baboon	1.1642	1.2345	1.4713	0.8464	0.7781	1.0736	0.9295	0.5629
Barbara	1.3153	1.3759	1.4573	0.8928	0.8888	1.4806	1.7143	1.4427
Boat	1.4497	1.3810	1.5806	0.8937	0.9573	1.7217	1.9830	1.6838
Cameraman	1.5780	1.7120	1.7682	0.9869	0.9276	1.6462	2.0755	2.3591
Jet	1.5864	1.5708	1.5082	0.9468	0.9629	1.7919	2.3645	2.5676
Lake	1.3265	1.4186	1.5917	0.9030	0.9355	1.6145	1.8513	1.6702
Peppers	1.1954	1.3616	1.5384	0.8841	0.9801	1.8626	2.4642	2.4403

(b) Entropy analysis

Image	Saeidi & Mirzakuchaki ²⁴		Our PQ-RDHEI		
	Share (avg)	Reconstructed Image	Share (avg)	Reconstructed Image	Original Image
Airplane	7.48	7.4887	7.9972	5.4474	5.4474
Baboon	7.48	7.3002	7.9991	7.3743	7.3743
Barbara	7.48	7.6029	7.9992	7.5674	7.5674
Boat	7.48	7.1996	7.9986	7.2095	7.2095
Cameraman	7.48	7.1063	7.9902	7.1045	7.1045
Jet	7.48	7.6184	7.9976	6.7240	6.7240
Lake	7.72	7.4887	7.9985	7.4758	7.4758
Peppers	7.48	7.6184	7.9983	7.5865	7.5865

Table 3. Embedding rates (bpp) for block size 4×4 - comparison of PQ-RDHEI with Saeidi & Mirzakuchaki²⁴

Image	Saeidi & Mirzakuchaki ²⁴								Our PQ-RDHEI							
	$\beta=3$, $T=2$	$\beta=3$, $T=3$	$\beta=4$, $T=2$	$\beta=4$, $T=3$	$\beta=5$, $T=2$	$\beta=5$, $T=3$	$\beta=6$, $T=2$	$\beta=6$, $T=3$	$\beta=3$, $T=2$	$\beta=3$, $T=3$	$\beta=4$, $T=2$	$\beta=4$, $T=3$	$\beta=5$, $T=2$	$\beta=5$, $T=3$	$\beta=6$, $T=2$	$\beta=6$, $T=3$
Airplane	0.3444	0.4327	0.3108	0.4845	0.3609	0.4138	0.2798	0.5908	0.9490	0.9677	1.8143	1.8702	1.9708	2.1614	1.5737	2.1313
Baboon	0.2203	0.2344	0.2500	0.2509	0.3589	0.3589	0.2803	0.5905	0.4030	0.5347	0.3794	0.6325	0.1298	0.3871	0.0122	0.0865
Barbara	0.2458	0.2870	0.2560	0.3108	0.3598	0.3642	0.2806	0.5905	0.6344	0.7357	0.9299	1.1325	0.7744	1.2262	0.2350	0.7354
Boat	0.2472	0.2874	0.2608	0.2846	0.3609	0.3716	0.2801	0.5915	0.7711	0.8465	1.0441	1.3791	0.7727	1.2732	0.3804	0.7779
Cameraman	0.2000	0.3503	0.3106	0.2922	0.3652	0.4160	0.2804	0.5921	0.7524	0.8202	1.2090	1.3778	1.4226	1.5795	1.2804	1.5129
Jet	0.2881	0.3399	0.2875	0.3398	0.3675	0.4029	0.2798	0.5948	0.8212	0.8742	1.3621	1.5489	1.4547	1.7579	1.0981	1.5221
Lake	0.2531	0.2995	0.2650	0.2922	0.3602	0.3779	0.2814	0.5908	0.6999	0.7944	1.0110	1.2666	0.8871	1.2962	0.5732	0.9647
Peppers	0.2400	0.2898	0.2517	0.2718	0.3587	0.3595	0.2803	0.5900	0.8726	0.9177	1.4456	1.6516	1.3640	1.9287	0.6633	1.3555

Regarding computational overhead, PQ-RDHEI incurs approximately 9% additional runtime compared to²⁴ (274ms vs. 251ms total pipeline for 512×512 images), primarily due to ML-KEM operations and Π_{LA} verification. Memory overhead remains modest at 11%, with both schemes exhibiting linear scaling in image size.

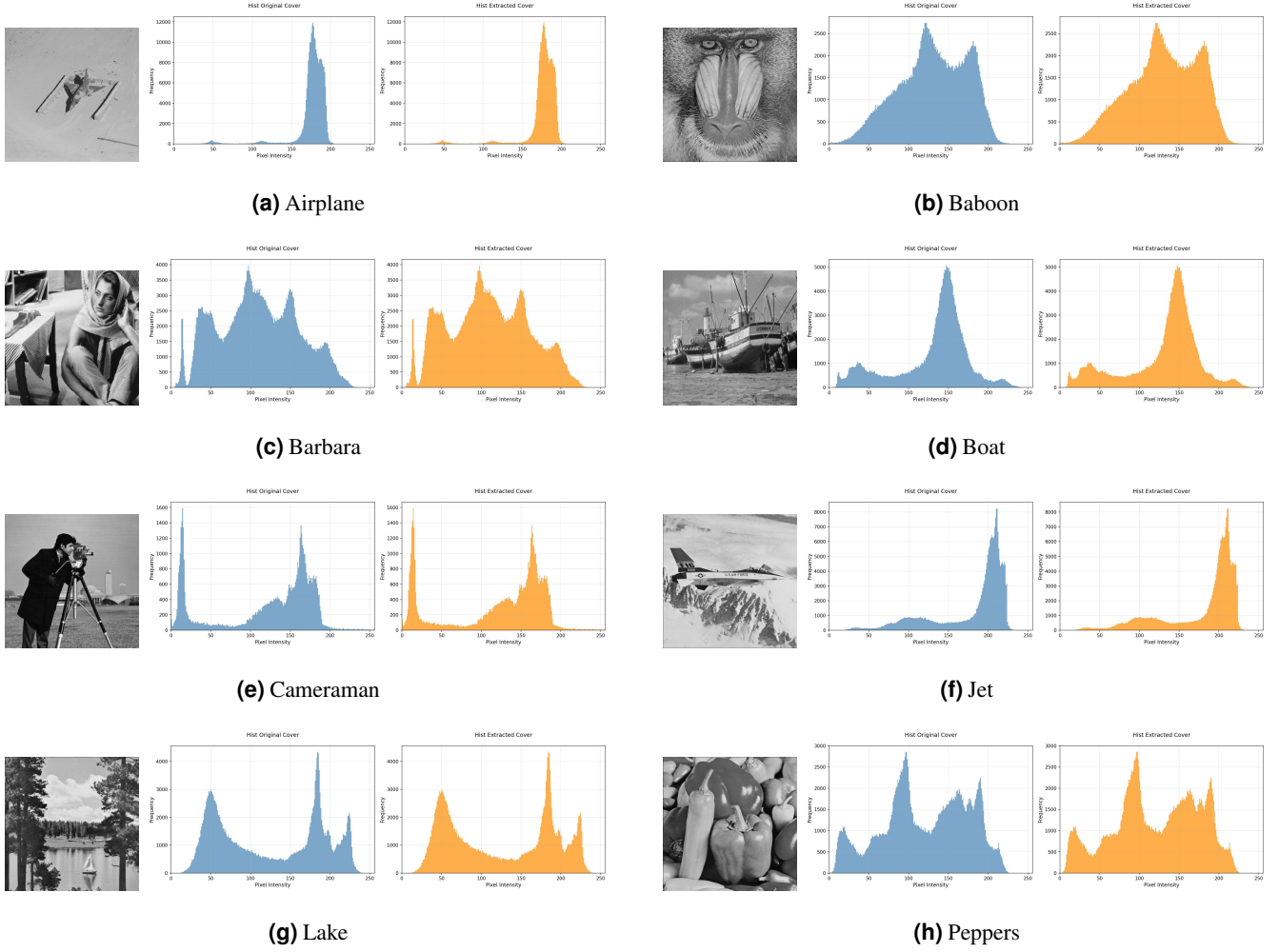


Figure 3. Histogram analysis of test images: original cover images (left) and their corresponding histogram comparisons (right) showing pixel distribution before and after extraction. The histogram comparisons demonstrate that the proposed PQ-RDHEI scheme achieves perfect reconstruction with pixel distributions of extracted images identical to the original images.

6 Security and Privacy Analysis of PQ-RDHED

Here, we formally address that our technique is correct (Theorems 1 and 2), compact (Theorem 6), semantically secure (Theorem 3), robust (Theorem 7), and provides payload privacy (Theorem 4) as well as separability (Theorem 5).

Remark 4 (Key Relationship and Security). For correct operation of PQ-RDHED, the hiding key $hk = K_h$ and extraction key $xk = K_x$ satisfy the constraint $K_h = K_x$. This enables the data hider to embed payloads that can be correctly extracted by the receiver. In the key distribution model: (ek, dk) are held by the content owner who encrypts images; (hk, xk) with $hk = xk$ are distributed such that the data hider receives hk for embedding and the receiver obtains xk for extraction. Under this constraint, payload privacy reduces to semantic security of the underlying encryption, as any party with hk can derive the masking keys. For applications requiring stronger payload privacy with $hk \neq xk$, the construction can be extended by including an additional public-key encryption layer for the payload under a key held only by the receiver.

Theorem 1. *Our proposed PQ-RDHED is correct (Definition 7); specifically, for all $\lambda \in \mathbb{N}$, $m \in \mathcal{M}$, $d \in \mathcal{D}$ with $|d| \leq \kappa(|m|)$:*

$$\Pr \left[\begin{array}{l} (ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda); c \leftarrow \text{Enc}(ek, m); \\ \hat{c} \leftarrow \text{Embed}(hk, c, d); (c', d') \leftarrow \text{Extract}(xk, \hat{c}) \end{array} : c' = c \wedge d' = d \right] = 1.$$

Proof. Let $(ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda)$ with $hk = K_h$ and $xk = K_x$ satisfying $K_h = K_x$ per Remark 4. Let $c = (c_{\text{kem}}, \{E'_k\}_{k=1}^\ell) \leftarrow$

$\text{Enc}(ek, m)$ and $\hat{c} = (c_{\text{kem}}, \{\hat{E}_k\}_{k=1}^\ell, \text{tag}) \leftarrow \text{Embed}(hk, c, d)$. Since $K_h = K_x$, the derived keys satisfy:

$$K_{\text{pos}}^{(\text{emb})} = H(K_h \| 0x02) = H(K_x \| 0x02) = K_{\text{pos}}^{(\text{ext})}, \quad K_{\text{mask}}^{(\text{emb})} = H(K_h \| 0x03) = H(K_x \| 0x03) = K_{\text{mask}}^{(\text{ext})}.$$

For each embeddable block $k \in \mathcal{E}$ with $\alpha_k \leq T$ distinct MSB types, define the encoding map $\phi_k : \{0, \dots, 2^\beta - 1\}^{n_b^2} \times \{0, 1\}^{\text{cap}_k} \rightarrow \{0, 1\}^{\beta \cdot n_b^2}$ by:

$$\phi_k \left(\{\text{msb}_{k,u}\}_{u=1}^{n_b^2}, \text{emb}_k \right) = \bigcup_{u=1}^{n_b^2} \left((0)^{p_{k,u}} \| 1 \| \overline{\text{code}_k(\text{msb}_{k,u})} \right),$$

where $\sum_{u=1}^{n_b^2} p_{k,u} = \text{cap}_k$ and $\bar{\cdot}$ denotes bit reversal. The delimiter bit ‘1’ ensures prefix-freeness, hence ϕ_k is injective. Since domain and codomain have equal cardinality $2^{\beta \cdot n_b^2}$, ϕ_k is bijective with inverse ϕ_k^{-1} computable in $O(n_b^2)$ time. The embedding constructs $\text{stream} = \text{len}(d) \| \tilde{d} \| \text{aux} \| \pi_{\text{share}}$ where $\tilde{d} = d \oplus \text{PRF}(K_{\text{mask}}, 0x00)[0 : |d|]$. Extraction recovers $\text{stream}' = \text{len}(d') \| \tilde{d}' \| \text{aux}' \| \pi'_{\text{share}}$ by applying ϕ_k^{-1} to each block $k \in \mathcal{E}'$ in lexicographic order. By bijectivity of ϕ_k and determinism of block ordering $\text{stream}' = \text{stream} \implies (\text{len}(d'), \tilde{d}', \text{aux}', \pi'_{\text{share}}) = (\text{len}(d), \tilde{d}, \text{aux}, \pi_{\text{share}})$. From the aforementioned steps of this proof, we have:

$$d' = \tilde{d}' \oplus \text{PRF}(K_{\text{mask}}^{(\text{ext})}, 0x00)[0 : |\tilde{d}'|] = \tilde{d} \oplus \text{PRF}(K_{\text{mask}}^{(\text{emb})}, 0x00)[0 : |\tilde{d}|] = d.$$

The auxiliary data $\text{aux} = L \| \mathcal{E} \| \{(\alpha_k, \{\text{msb}_k^{(j)}\}_{j=1}^{\alpha_k})_{k \in \mathcal{E}}\}$ encodes the location map and original MSB metadata. For each k with $L[k] = 1$, reconstruction applies: $E'_k[u] = (\text{msb}_{k,u} \cdot 2^{8-\beta}) + (\hat{E}_k[u] \bmod 2^{8-\beta})$, $\forall u \in [n_b^2]$. Since $\text{msb}_{k,u}$ is recovered exactly via ϕ_k^{-1} and LSB positions are unmodified by embedding, $E'_k = E_k$ for all $k \in [\ell]$. Thus $c' = (c_{\text{kem}}, \{E'_k\}_{k=1}^\ell) = c$. $\square$

Theorem 2. PQ-RDHED satisfies decryption correctness (Definition 8). For all $\lambda \in \mathbb{N}$, $m \in \mathcal{M}$:

$$\Pr \left[(ek, dk, hk, xk) \leftarrow \text{KeyGen}(1^\lambda); c \leftarrow \text{Enc}(ek, m) : \text{Dec}(dk, c) = m \right] \geq 1 - \delta_{\text{kem}},$$

where $\delta_{\text{kem}} = \text{negl}(\lambda)$ denotes the decapsulation failure probability of ML-KEM.

Proof Sketch. Let $(pk, sk) \leftarrow \text{ML-KEM.KeyGen}(1^\lambda)$ with $ek = pk$, $dk = sk$. Let $c = (c_{\text{kem}}, \{E'_k\}_{k=1}^\ell) \leftarrow \text{Enc}(ek, m)$ where $(c_{\text{kem}}, K) \leftarrow \text{ML-KEM.Encaps}(pk)$. By the $(1 - \delta_{\text{kem}})$ -correctness of ML-KEM: $\Pr[K' = K \mid K' \leftarrow \text{ML-KEM.Decaps}(sk, c_{\text{kem}})] \geq 1 - \delta_{\text{kem}}$. Define event $\mathcal{E}_{\text{kem}} := \{K' = K\}$. We condition on $\mathcal{E}_{\text{kem}}$ for the remainder. Conditioned on $\mathcal{E}_{\text{kem}}$, Dec computes $K'_e = H(K' \| 0x00) = H(K \| 0x00) = K_e$, $K'_\pi = H(K' \| 0x01) = H(K \| 0x01) = K_\pi$. Let $\sigma = \text{FisherYates}(\ell, K_\pi) : [\ell] \rightarrow [\ell]$. Since FisherYates is deterministic given (ℓ, K_π) and produces a bijection, σ^{-1} exists and is computable. By construction: $E'_{\sigma(k)} = E_k$ during encryption, hence $E_k = E'_{\sigma^{-1}(k)} = E'_k \circ \sigma^{-1}$. For each $k \in [\ell]$, let $E_k = \{e_{k,u}\}_{u=1}^{n_b^2}$ with $e_{k,u} = p_{k,u} \oplus \mu_k$ where $\mu_k = \text{PRF}(K_e, k)[0 : 8]$. Decryption computes $\mu'_k = \text{PRF}(K'_e, k)[0 : 8] = \text{PRF}(K_e, k)[0 : 8] = \mu_k$ and $p'_{k,u} = e_{k,u} \oplus \mu'_k = (p_{k,u} \oplus \mu_k) \oplus \mu_k = p_{k,u}$, $\forall u \in [n_b^2]$. The partition $\{B_k\}_{k=1}^\ell$ of m into $n_b \times n_b$ blocks is deterministic given (W, H, n_b) . Since $B'_k = \{p'_{k,u}\}_{u=1}^{n_b^2} = \{p_{k,u}\}_{u=1}^{n_b^2} = B_k$ for all k , reconstruction yields $m' = m$. Eventually, we will have the probability bound:

$$\Pr[\text{Dec}(dk, c) = m] \geq \Pr[\mathcal{E}_{\text{kem}}] \cdot \Pr[\text{Dec}(dk, c) = m \mid \mathcal{E}_{\text{kem}}] = (1 - \delta_{\text{kem}}) \cdot 1 = 1 - \delta_{\text{kem}}.$$

$\square$

Theorem 3. PQ-RDHED satisfies semantic security (Definition 9) in the QROM. For any QPT adversary $\mathcal{A}$ making at most q_H quantum queries to H and q_D classical decapsulation queries: $\text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{sem}}(\lambda) \leq \text{Adv}_{\text{ML-KEM}, \mathcal{B}_1}^{\text{IND-CCA}}(\lambda) + \sum_{k=1}^\ell \text{Adv}_{\text{PRF}, \mathcal{B}_{2,k}}^{\text{PRF}}(\lambda)$, where $\mathcal{B}_1$ is a QPT reduction to IND-CCA security of ML-KEM and $\{\mathcal{B}_{2,k}\}_{k=1}^\ell$ are QPT reductions to PRF security.

Proof. We proceed via a sequence of games $G_0, G_1, \dots, G_{\ell+1}$.

Game G_0 : Define $p_0 := \Pr[G_0 \text{ outputs } 1]$. The real experiment $\text{Expt}_{\mathcal{A}, b}^{\text{sem}}(\lambda)$:

1. $(pk, sk) \leftarrow \text{ML-KEM.KeyGen}(1^\lambda)$; $K_h \leftarrow \{0, 1\}^\lambda$; $K_x \leftarrow \{0, 1\}^\lambda$.
2. $\mathcal{A}(pk, K_h, K_x)$ outputs (m_0, m_1) with $|m_0| = |m_1|$.
3. $(c_{\text{kem}}, K) \leftarrow \text{ML-KEM.Encaps}(pk)$; $K_e \leftarrow H(K \| 0x00)$; $K_\pi \leftarrow H(K \| 0x01)$.
4. For $k \in [\ell]$: $\mu_k \leftarrow \text{PRF}(K_e, k)[0 : 8]$; $E_k \leftarrow \{m_b[k, u] \oplus \mu_k\}_{u=1}^{n_b^2}$.
5. $\sigma \leftarrow \text{FisherYates}(\ell, K_\pi)$; $c^* \leftarrow (c_{\text{kem}}, \{E_{\sigma^{-1}(k)}\}_{k=1}^\ell)$.

6. $b' \leftarrow \mathcal{A}(c^*)$; output b' .

Game G_1 : Identical to G_0 except: replace $(c_{\text{kem}}, K) \leftarrow \text{ML-KEM.Encaps}(pk)$ with $(c_{\text{kem}}, K) \leftarrow \text{ML-KEM.Encaps}(pk)$ but substitute $K \leftarrow \{0,1\}^{256}$ uniformly at random (i.e., K is independent of c_{kem}). Define $p_1 := \Pr[G_1 \text{ outputs } 1]$.

Claim 1: $|p_0 - p_1| \leq \text{Adv}_{\text{ML-KEM}, \mathcal{B}_1}^{\text{IND-CCA}}(\lambda)$.

Proof of Claim 1: Construct $\mathcal{B}_1$ as follows: $\mathcal{B}_1$ receives $(pk, c_{\text{kem}}^*, K^*)$ from the IND-CCA challenger where K^* is either the real encapsulated key or uniform random. $\mathcal{B}_1$ sets $c_{\text{kem}} := c_{\text{kem}}^*$, $K := K^*$, simulates the remainder of the game, and outputs $\mathcal{A}$'s guess. If K^* is real, $\mathcal{B}_1$ simulates G_0 ; if K^* is random, $\mathcal{B}_1$ simulates G_1 . Thus:

$$|p_0 - p_1| = \left| \Pr[\mathcal{B}_1^{\mathcal{O}}(\text{real}) = 1] - \Pr[\mathcal{B}_1^{\mathcal{O}}(\text{rand}) = 1] \right| = \text{Adv}_{\text{ML-KEM}, \mathcal{B}_1}^{\text{IND-CCA}}(\lambda).$$

Games G_{1+k} for $k \in [\ell]$: Identical to G_k except: replace $\mu_k = \text{PRF}(K_e, k)[0:8]$ with $\mu_k \leftarrow \{0, \dots, 255\}$ sampled uniformly at random. Define $p_{1+k} := \Pr[G_{1+k} \text{ outputs } 1]$.

Claim 2: For each $k \in [\ell]$: $|p_k - p_{1+k}| \leq \text{Adv}_{\text{PRF}, \mathcal{B}_{2,k}}^{\text{PRF}}(\lambda)$.

Proof of Claim 2: Construct $\mathcal{B}_{2,k}$ with oracle access to $\mathcal{O}$ (either $\text{PRF}(K_e, \cdot)$ or a random function). For indices $j < k$, $\mathcal{B}_{2,k}$ samples $\mu_j \leftarrow \{0, \dots, 255\}$. For index k , $\mathcal{B}_{2,k}$ queries $\mu_k \leftarrow \mathcal{O}(k)[0:8]$. For indices $j > k$, $\mathcal{B}_{2,k}$ queries $\mu_j \leftarrow \mathcal{O}(j)[0:8]$. If $\mathcal{O} = \text{PRF}(K_e, \cdot)$, this simulates G_k ; if $\mathcal{O}$ is random at position k , this simulates G_{1+k} . Thus $|p_k - p_{1+k}| \leq \text{Adv}_{\text{PRF}, \mathcal{B}_{2,k}}^{\text{PRF}}(\lambda)$.

To analyze $G_{\ell+1}$, all masks $\{\mu_k\}_{k=1}^\ell$ are uniformly random and independent. For each block k and pixel position u : $E_k[u] = m_b[k, u] \oplus \mu_k$, where $\mu_k \leftarrow \{0, \dots, 255\}$ is independent of m_b . By the XOR-lemma, $E_k[u]$ is uniformly distributed over $\{0, \dots, 255\}$ independent of $m_b[k, u]$. Since the permutation σ is determined by $K_\pi = H(K \| 0x01)$ with K uniform random (independent of b), the permuted blocks $\{E_{\sigma^{-1}(k)}\}_{k=1}^\ell$ are identically distributed for $b=0$ and $b=1$. Hence:

$$p_{\ell+1} = \Pr[G_{\ell+1} \text{ outputs } 1 \mid b=0] = \Pr[G_{\ell+1} \text{ outputs } 1 \mid b=1] \implies \Pr[b' = b] = \frac{1}{2}.$$

To calculate the advantage, we will have:

$$\begin{aligned} \text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{sem}}(\lambda) &= |\Pr[b' = 1 \mid b=0] - \Pr[b' = 1 \mid b=1]| = 2 \left| p_0 - \frac{1}{2} \right| \leq 2 \left(|p_0 - p_1| + \sum_{k=1}^\ell |p_k - p_{1+k}| + \left| p_{\ell+1} - \frac{1}{2} \right| \right) \\ &\leq 2 \left(\text{Adv}_{\text{ML-KEM}, \mathcal{B}_1}^{\text{IND-CCA}}(\lambda) + \sum_{k=1}^\ell \text{Adv}_{\text{PRF}, \mathcal{B}_{2,k}}^{\text{PRF}}(\lambda) + 0 \right) = 2 \cdot \text{Adv}_{\text{ML-KEM}, \mathcal{B}_1}^{\text{IND-CCA}}(\lambda) + 2 \sum_{k=1}^\ell \text{Adv}_{\text{PRF}, \mathcal{B}_{2,k}}^{\text{PRF}}(\lambda). \end{aligned}$$

□

Theorem 4. PQ-RDHED satisfies payload privacy (Definition 10) in the QROM under the extended key model where $hk \neq xk$. For any QPT adversary $\mathcal{A}$ we have $\text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{pay}}(\lambda) \leq 2 \cdot \text{Adv}_{\text{PRF}, \mathcal{B}}^{\text{PRF}}(\lambda)$. Under the standard key model with $hk = xk$ (Remark 4), payload privacy reduces to semantic security with $\text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{pay}}(\lambda) \leq \text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{sem}}(\lambda)$.

Proof. Case 1 (Extended Key Model, $hk \neq xk$): Consider the experiment $\text{Expt}_{\mathcal{A}, b}^{\text{pay}}(\lambda)$ where $\mathcal{A}$ receives (ek, hk) but not xk .

Game G_0 : The real payload privacy experiment. The challenger computes $\tilde{d} = d_b \oplus \text{PRF}(K_{\text{mask}}, 0x00)[0 : |d_b|]$ where $K_{\text{mask}} = H(K_h \| 0x03)$.

Game G_1 : Replace $\text{PRF}(K_{\text{mask}}, 0x00)[0 : |d_b|]$ with $r \leftarrow \{0,1\}^{|d_b|}$ sampled uniformly.

Claim: $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq \text{Adv}_{\text{PRF}, \mathcal{B}}^{\text{PRF}}(\lambda)$.

Proof: Construct $\mathcal{B}$ with oracle $\mathcal{O}$ (either $\text{PRF}(K_{\text{mask}}, \cdot)$ or random). $\mathcal{B}$ simulates the experiment, querying $\mathcal{O}(0x00)$ to obtain the mask. Distinguishing G_0 from G_1 implies distinguishing PRF from random. In G_1 , $\tilde{d} = d_b \oplus r$ where r is uniform random. For any d_0, d_1 with $|d_0| = |d_1|$: $\tilde{d}_0 = d_0 \oplus r \stackrel{d}{=} r \stackrel{d}{=} d_1 \oplus r = \tilde{d}_1$, where $\stackrel{d}{=}$ denotes identical distribution. Thus $\Pr[G_1 = 1 \mid b=0] = \Pr[G_1 = 1 \mid b=1]$, implying $\text{Adv}_{G_1}^{\text{pay}} = 0$. For Case 1, the advantage bound will be:

$$\begin{aligned} \text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{pay}}(\lambda) &= |\Pr[G_0 = 1 \mid b=0] - \Pr[G_0 = 1 \mid b=1]| \\ &\leq |\Pr[G_0 = 1] - \Pr[G_1 = 1]| + |\Pr[G_1 = 1 \mid b=0] - \Pr[G_1 = 1 \mid b=1]| \\ &\quad + |\Pr[G_1 = 1] - \Pr[G_0 = 1]| \leq 2 \cdot \text{Adv}_{\text{PRF}, \mathcal{B}}^{\text{PRF}}(\lambda) + 0. \end{aligned}$$

Case 2 (Standard Key Model, $hk = xk$): When $\mathcal{A}$ possesses $hk = K_h = K_x = xk$, it can derive K_{mask} and unmask $\tilde{d}$ to recover d_b . In this setting, payload privacy is implied by semantic security: if $\mathcal{A}$ could distinguish payloads, construct $\mathcal{A}'$ that embeds d_0, d_1 into encryptions of distinct messages and uses $\mathcal{A}$'s output to distinguish ciphertexts. Thus $\text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{pay}} \leq \text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{sem}}$. □

Theorem 5. *The proposed PQ-RDHED scheme (Definition 16) satisfies the property of separability (Definition 11).*

Proof. We verify both conditions of Definition 11.

Condition (I) - Extraction-First: We show $\text{Extract}(xk, \cdot)$ is computable without dk . Examining Extract in Definition 16:

- Step 1 derives $(K_{\text{pos}}, K_{\text{mask}})$ from $xk = K_x$ via H ; no dependence on $dk = sk$.
- Steps 2–4 perform MSB analysis and Huffman decoding on encrypted blocks $\{\hat{E}_k\}$; operations are combinatorial on ciphertext.
- Steps 5–8 parse, verify, and unmask the payload using K_{mask} ; verification uses Π_{LA} . VerifyShare which requires only π_{share} .
- Steps 9–11 reconstruct c from aux metadata; no decryption invoked.

Thus $\text{Extract}: \{0,1\}^\lambda \times \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{D}$ depends only on $(xk, \hat{c})$.

Condition (II) - Decryption-First: Define $\text{Dec}': \{0,1\}^* \times \mathcal{C} \rightarrow \mathcal{M}$ as follows: Parse $\hat{c} = (c_{\text{kem}}, \{\hat{E}_k\}_{k=1}^\ell, \text{tag})$. Compute $K \leftarrow \text{ML-KEM.Decaps}(sk, c_{\text{kem}})$. Derive $K_e = H(K \| 0x00)$, $K_\pi = H(K \| 0x01)$. Compute $\sigma^{-1} = \text{FisherYates}(\ell, K_\pi)^{-1}$. For each $k \in [\ell]$: set $\hat{E}'_k = \hat{E}_{\sigma(k)}$; compute $\mu_k = \text{PRF}(K_e, k)[0:8]$; for each $u \in [n_b^2]$: $\hat{p}_{k,u} = \hat{E}'_k[u] \oplus \mu_k$. Output $\hat{m} = \{\hat{p}_{k,u}\}_{k \in [\ell], u \in [n_b^2]}$. The output $\hat{m}$ encodes: (i) the original image m in the $(8-\beta)$ LSB positions of each pixel; (ii) the embedded stream (Huffman-encoded MSBs interleaved with payload bits) in the β MSB positions of pixels in embeddable blocks. Formally, for embeddable block k and pixel u :

$$\hat{p}_{k,u} = \underbrace{(0)^{p_{k,u}} \| 1 \| \text{code}_k(\text{msb}_{k,u})}_{\beta \text{ MSBs: encoded data}} \| \underbrace{p_{k,u}}_{(8-\beta) \text{ LSBs: original}} \pmod{2^{8-\beta}}.$$

Recovery of (m, d) from $\hat{m}$ requires only the location map L (identifiable via encoding structure) and Huffman decoding tables (reconstructible from α_k counts), both extractable from $\hat{m}$ without xk . $\square$

Theorem 6. *The proposed PQ-RDHED scheme (Definition 16) satisfies property of compactness (Definition 12) with overhead $p(\lambda, |d|) = \lambda$. Specifically, for all valid (c, d) we will have $|\text{Embed}(hk, c, d)| = |c| + \lambda$.*

Proof Sketch. Let $c = (c_{\text{kem}}, \{E'_k\}_{k=1}^\ell)$ and $\hat{c} = (c_{\text{kem}}, \{\hat{E}_k\}_{k=1}^\ell, \text{tag})$. To consider component sizes, $|c_{\text{kem}}|$ is fixed by ML-KEM parameters; identical in c and $\hat{c}$, $|\{E'_k\}_{k=1}^\ell|$ is $\ell \cdot n_b^2 \cdot 8$ bits (each pixel is 8 bits), $|\{\hat{E}_k\}_{k=1}^\ell|$ is $\ell \cdot n_b^2 \cdot 8$ bits. Embedding modifies only the β -MSB positions within existing pixels; no expansion occurs since Huffman encoding plus payload occupies exactly $\beta \cdot n_b^2$ bits per embeddable block (the same space as the original MSBs); and $|\text{tag}|$ is λ bits (output length of H). To compute the total sizes here, $|c| = |c_{\text{kem}}| + \ell \cdot n_b^2 \cdot 8$, $|\hat{c}| = |c_{\text{kem}}| + \ell \cdot n_b^2 \cdot 8 + \lambda = |c| + \lambda$. Thus $|\hat{c}| \leq |c| + p(\lambda, |d|)$ with $p(\lambda, |d|) = \lambda$, satisfying Definition 12. $\square$

Theorem 7. *PQ-RDHED satisfies embedding robustness (Definition 13) in the QROM. For any QPT adversary $\mathcal{A}$ making at most q_H quantum queries to H we have $\text{Adv}_{\text{PQ-RDHED}, \mathcal{A}}^{\text{robust}}(\lambda) \leq \text{Adv}_{H, \mathcal{B}_1}^{\text{CR}}(\lambda) + \text{Adv}_{\Pi_{\text{LA}}, \mathcal{B}_2}^{\text{bind}}(\lambda) + \frac{q_H^2}{2\lambda}$.*

Proof. Consider $\mathcal{A}$ outputting $(c, d, \hat{c}^*)$ such that $\hat{c}^* \neq \text{Embed}(hk, c, d)$ but $\text{Extract}(xk, \hat{c}^*) = (c, d)$. Let $\hat{c} := \text{Embed}(hk, c, d) = (c_{\text{kem}}, \{\hat{E}_k\}, \text{tag})$ be the honest embedding, and parse $\hat{c}^* = (c_{\text{kem}}^*, \{\hat{E}_k^*\}, \text{tag}^*)$.

Case 1: $c_{\text{kem}}^* \neq c_{\text{kem}}$. The extraction algorithm verifies $\text{tag}^* = H(K_x \| c_{\text{kem}}^* \| H(\text{stream}'))$. For extraction to yield $c = (c_{\text{kem}}, \cdot)$ from $\hat{c}^* = (c_{\text{kem}}^*, \cdot)$ with $c_{\text{kem}}^* \neq c_{\text{kem}}$, we require the recovered ciphertext to contain c_{kem} . However, Extract outputs $c' = (c_{\text{kem}}^*, \{E'_k\})$, preserving c_{kem}^* . Thus $c' \neq c$, contradiction.

Case 2: $c_{\text{kem}}^* = c_{\text{kem}}$ and $\{\hat{E}_k^*\} \neq \{\hat{E}_k\}$. Let stream and stream' denote the embedded streams for $\hat{c}$ and $\hat{c}^*$ respectively. Since $\text{Extract}(xk, \hat{c}^*) = (c, d)$, then the payload satisfies $d' = d$ (where $d' = \tilde{d}' \oplus \text{PRF}(K_{\text{mask}}, 0x00)[0:|d'|]$), the auxiliary data satisfies $\text{aux}' \rightarrow c$ (reconstruction yields original ciphertext), and the VSS satisfies $\Pi_{\text{LA}}.\text{Reconstruct}(\cdot, \pi_{\text{share}}') = H(\text{aux}' \| d)$.

Subcase 2a: $\text{stream}' = \text{stream}$. Since embedding is deterministic given (hk, c, stream) , and $\text{stream}' = \text{stream}$ implies identical embedding bits, we have $\{\hat{E}_k^*\} = \{\hat{E}_k\}$, contradicting the assumption.

Subcase 2b: $\text{stream}' \neq \text{stream}$ with $(\text{aux}', d', \pi_{\text{share}}') \neq (\text{aux}, d, \pi_{\text{share}})$.

Subcase 2b-i: If $(\text{aux}', d') \neq (\text{aux}, d)$ but $H(\text{aux}' \| d') = H(\text{aux} \| d)$ (required for VSS verification against the same commitment), then $(\text{aux}' \| d', \text{aux} \| d)$ constitutes a collision in H . Construct $\mathcal{B}_1$: run $\mathcal{A}$ to obtain $(c, d, \hat{c}^*)$; compute honest embedding $\hat{c}$; extract stream' from $\hat{c}^*$; if $\text{aux}' \| d' \neq \text{aux} \| d$ and both hash to the same value, output the collision pair. Thus: $\Pr[\text{Subcase 2b-i}] \leq \text{Adv}_{H, \mathcal{B}_1}^{\text{CR}}(\lambda)$.

Subcase 2b-ii: If $\pi_{\text{share}}' \neq \pi_{\text{share}}$ but both verify and reconstruct to $H(\text{aux} \| d)$, this violates the binding property of Π_{LA} . Construct $\mathcal{B}_2$: run $\mathcal{A}$; extract both π s; if distinct π s reconstruct to the same value, output the binding break. Thus: $\Pr[\text{Subcase 2b-ii}] \leq \text{Adv}_{\Pi_{\text{LA}}, \mathcal{B}_2}^{\text{bind}}(\lambda)$.

Case 3: $\text{tag}^* \neq \text{tag}$ with $(c_{\text{kem}}^*, \{\hat{E}_k^*\}) = (c_{\text{kem}}, \{\hat{E}_k\})$. Then $\hat{c}^* = (c_{\text{kem}}, \{\hat{E}_k\}, \text{tag}^*)$ with $\text{tag}^* \neq \text{tag}$. The extraction verifies $\text{tag}^* = H(K_x \| c_{\text{kem}} \| H(\text{stream}'))$. Since embedded blocks are identical, $\text{stream}' = \text{stream}$, hence the correct tag is

$\text{tag} = H(K_x \| c_{\text{kem}} \| H(\text{stream}))$. If $\text{tag}^* \neq \text{tag}$, verification fails and extraction outputs $\perp \neq (c, d)$, contradiction. Finally, for the total advantage we will have:

$$\text{Adv}_{\text{PQ-RDHEI}, \mathcal{A}}^{\text{robust}}(\lambda) \leq \Pr[\text{Case 2b-i}] + \Pr[\text{Case 2b-ii}] + \Pr[\text{QROM collision}] \leq \text{Adv}_{H, \mathcal{B}_1}^{\text{CR}}(\lambda) + \text{Adv}_{\Pi_{\text{LA}}, \mathcal{B}_2}^{\text{bind}}(\lambda) + \frac{q_H^2}{2\lambda}.$$

□

Under the quantum hardness of $\text{MLWE}_{n,k,q,\chi}$, post-quantum collision resistance of H in the QROM, and quantum-secure instantiations of PRF (via SHAKE256) and Π_{LA} (hash-based VSS), all security properties established in Theorems 1–7 hold against QPT adversaries. Concretely, for any QPT adversary $\mathcal{A}$ and security parameter λ :

$$\text{Adv}_{\mathcal{A}}^{\text{sem}}(\lambda), \text{Adv}_{\mathcal{A}}^{\text{pay}}(\lambda), \text{Adv}_{\mathcal{A}}^{\text{unlink}}(\lambda), \text{Adv}_{\mathcal{A}}^{\text{robust}}(\lambda) = \text{negl}(\lambda).$$

The reductions in Theorems 3, 4, and 7 are polynomial-time and preserve quantum query complexity. By³², $\text{Adv}_{\text{ML-KEM}}^{\text{IND-CCA}}(\lambda) = \text{negl}(\lambda)$ under $\text{MLWE}_{n,k,q,\chi}$ in the QROM. By³³, SHAKE256 instantiates a quantum-secure PRF with $\text{Adv}_{\text{PRF}}^{\text{PRF}}(\lambda) = \text{negl}(\lambda)$. By³⁴, SHA3-256 provides $\text{Adv}_H^{\text{CR}}(\lambda) = \text{negl}(\lambda)$ against quantum adversaries. The VSS scheme Π_{LA} inherits post-quantum binding from its hash-based commitments³⁵, yielding $\text{Adv}_{\Pi_{\text{LA}}}^{\text{bind}}(\lambda) = \text{negl}(\lambda)$. Substituting into the advantage bounds:

$$\text{Adv}_{\mathcal{A}}^{\text{sem}}(\lambda) \leq 2 \cdot \text{negl}(\lambda) + 2\ell \cdot \text{negl}(\lambda) = \text{negl}(\lambda), \text{Adv}_{\mathcal{A}}^{\text{robust}}(\lambda) \leq \text{negl}(\lambda) + \text{negl}(\lambda) + 2^{-\Omega(\lambda)} = \text{negl}(\lambda).$$

Analogous substitutions yield negligible advantages for payload privacy and unlinkability.

7 Conclusion

This work presents the first formal cryptographic treatment of Reversible Data Hiding in Encrypted Images. We have defined RDHEI as a cryptographic primitive with well-defined syntax, accurate correctness conditions and security traits such as semantic security, payload privacy, separability, compactness, and robustness. Following this formalization, PQ-RDHEI was developed, which is post-quantum secure under the MLWE assumption in the quantum random oracle model. The construction relies on ML-KEM for key encapsulation, pseudorandom functions realized through SHAKE256, the SHA3-256 hash function, and hash-based verifiable secret sharing. Our experimental results show that PQ-RDHEI not only allows for greater embedding capacity than previous methods but also provides formal security guarantees that were not available in the literature before.

As future work, one direction can be focusing on the efficiency of RDHEI by a more formal approach; e.g., apart from compactness, we can say an RDHEI scheme has α -capacity efficiency for $\alpha \in (0, 1]$ if $\kappa(n) \geq \alpha \cdot n$ for all $n \in \mathbb{N}$, and this definition can have further development. Another future direction can be improving our PQ-RDHEI to achieve the strong variant of compactness (Definition 12); strong compactness can be achieved in PQ-RDHEI by omitting tag at the cost of robustness guarantees, but without omitting that it is not possible due to the λ -bit authentication tag. Considering the limitation of RDHEI methods in gray-scale, extending the formalization to color images presents additional research opportunities. Moreover, we aim to eliminate the secret sharing phase in RDHEI totally via silent-setup threshold cryptosystems³⁸ which boosts general efficiency; however, current silent-setup constructions rely on witness encryption, which remains impractical and quantum-vulnerable^{39–41}, so designing a post-quantum silent-setup scheme supporting homomorphic operations to get used in PQ-RDHEI is another open problem.

8 Data Availability

All implementation details are available from the author on request. The data used in this study are standard benchmark images obtained from the publicly available USC-SIPI Image Database available at: <http://sipi.usc.edu/database>.

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10 Author Contributions

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The authors declare no competing interests.

12 Ethics Approval and Consent to Participate

The author confirms the sole responsibility for this manuscript.

13 Consent for Publication

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14 Additional Information

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